



**Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 2 • STANDARD 6.DS.5

# Represent and Describe Data in a Box Plot

Lesson 2-4 • Teacher Copy

**Name:** \_\_\_\_\_

**Date:** \_\_\_\_\_

**Class:** \_\_\_\_\_



## TODAY'S OBJECTIVES

**Content Objective:** I can make and read a box plot to summarize a data set.

**Language Objective:** I can explain my box plot using the words box plot, median, quartile, and interquartile range.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



**Fill in each blank as we go. Use the Word Bank to help you.**



## WORD BANK – FILL EACH BLANK WITH THE BEST WORD

Display Data with Box Plots

Box Plot

Median

Quartile

Interquartile Range

Data distribution

Variability



Tap any word to see what it means and a picture.

1 A box plot shows the  -number summary: least, Q1, median, Q3, greatest.

2 A graph that shows data with a box and whiskers based on the five-number summary is a .

3 The middle value of the data, shown as a line inside the box, is the .

4 A value that divides the data into four equal parts is a .

5 The distance between the first and third quartiles, showing the middle 50% of data, is the .

6 The way data values are spread out is the  .

7 How spread out the data values are is the  .

## Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

### 1. Understand the Question

EXPLAIN

**Which team is more consistently scoring high? Use the box plot values to support your answer?**

**Your job:** Answer the question. Use math evidence. Explain how the evidence proves your answer.

*Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.*

## 2. Plan Your Math Words

Check at least two words you will use.

- ☐ **Display Data with Box Plots** — Show a data set's five-number summary in a box-and-whisker display.  
*Representar datos con diagramas de caja — Mostrar el resumen de cinco números de un conjunto de datos en un diagrama de caja y bigotes.*

- ☐ **Box Plot** — A graph that shows how data is spread using five key numbers.  
*Diagrama de caja — Una gráfica que muestra cómo se reparten los datos con cinco números clave.*

- ☐ **Median** — The middle number when you put them in order.  
*Mediana — El número del medio cuando los pones en orden.*

- ☐ **Quartile** — Numbers that split the data into four equal parts.  
*Cuartil — Números que dividen los datos en cuatro partes iguales.*

- ☐ **Interquartile Range** — The spread of the middle half of the data ( $Q_3 - Q_1$ ).  
*Rango intercuartílico — La extensión de la mitad central de los datos ( $Q_3 - Q_1$ ).*

**Say it first:** Say your idea to a partner or quietly to yourself before you write.

*Di tu idea a un compañero o en voz baja antes de escribir.*

**Team A's IQR is \_\_\_\_ and Team B's IQR is \_\_\_\_.** **Team B's middle 50% scores between \_\_\_\_ and \_\_\_\_.** **The fan is wrong because \_\_\_\_.** **The box plot shows that Team B is more consistent because its  $Q_1$  and median are higher and its IQR (10) is smaller, while \_\_\_\_.**

*Di tu respuesta primero: Mi respuesta es \_\_\_\_ porque \_\_\_\_.*

### 3. Build Your Explanation

**Model the parts:** This model shows the parts of an explanation. It does not answer your writing question.

**Claim:** A box plot is built from the minimum, Q1, median, Q3, and maximum.

**Evidence:** Students name the five-number summary (min, Q1, median, Q3, max) and find the median (14) for the 11 ordered scores.

**Reasoning:** This evidence connects the definition of box plot to the mathematical claim.

#### Start

Most language support

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- Team A's IQR is \_\_\_\_ and Team B's IQR is \_\_\_\_ . Team B's middle 50% scores between \_\_\_\_ and \_\_\_\_ . The fan is wrong because \_\_\_\_ . The box plot shows that Team B is more consistent because its Q1 and median are higher and its IQR (10) is smaller, while \_\_\_\_ .

- My answer is \_\_\_\_ .

*Mi respuesta es \_\_\_\_ .*

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#### Build

Some language support

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- My claim is \_\_\_\_ .

*Mi afirmación es \_\_\_\_ .*

- I know because \_\_\_\_ .

*Lo sé porque \_\_\_\_ .*

- So \_\_\_\_ .

*Entonces \_\_\_\_ .*

## Explain

### Light language support

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*
- **My math evidence is \_\_\_\_.**  
*Mi evidencia matemática es \_\_\_\_.*
- **This proves \_\_\_\_ because \_\_\_\_.**  
*Esto demuestra \_\_\_\_ porque \_\_\_\_.*

## 4. Check Your Explanation

- ☐ I answered the question.  
*Respondí la pregunta.*
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.  
*Usé evidencia matemática: números, una ecuación, un modelo o una comparación.*
- ☐ I explained how the evidence proves my answer.  
*Explicué cómo la evidencia demuestra mi respuesta.*
- ☐ I used at least two math words.  
*Usé por lo menos dos palabras matemáticas.*
- ☐ I reread complete sentences and punctuation.  
*Volví a leer mis oraciones completas y la puntuación.*

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## Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.



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## Answer Key & Teacher Guide

1. **Notes 1:** Display Data with Box Plots
2. **Notes 2:** Box Plot
3. **Notes 3:** Median
4. **Notes 4:** Quartile
5. **Notes 5:** Interquartile Range
6. **Notes 6:** Data distribution
7. **Notes 7:** Variability
8. **Watch for this mistake:** *A common mistake in Display Data: Box Plots is treating a quartile as just one data point instead of the median of a half — for the lower half 4, 8, 10, 14, some students pick 10 (the last value) as Q1 instead of finding the median of that half:  $Q1 = (8 + 10) \div 2 = 9$ . A second common slip is finding IQR by adding the quartiles instead of subtracting, writing  $IQR = Q3 + Q1$  (like  $30 + 18 = 48$ ) instead of  $IQR = Q3 - Q1$  ( $30 - 18 = 12$ ) — always order the data first, then subtract Q1 from Q3 to find the spread of the middle 50%.*
9. **Try It 1:** A.  $Q1 = 12$ ,  $Q3 = 28$  — *With 8 values the median splits the data into a lower half (6, 10, 14, 18) and an upper half (22, 26, 30, 34). Q1 is the median of the lower half:  $(10 + 14) \div 2 = 12$ . Q3 is the median of the upper half:  $(26 + 30) \div 2 = 28$ .*
10. **Try It 2:** A. Between 15 and 30 — *The box of a box plot stretches from Q1 to Q3, and that box always holds the middle 50% of the data. Here the box goes from  $Q1 = 15$  to  $Q3 = 30$ .*